

1 Solution — GIAM 1.4

Problem 1

1.1 Problem

An integer n is *doubly-even* if it is even, and the integer m guaranteed to exist because n is even is itself even. Is 0 doubly-even? What are the first 3 positive, doubly-even integers?

1.2 Unpacking the Definition

Section 1.4 defines: n is **even** exactly when there is an integer m with $n = 2m$. That m is the “half” of n . The new term *doubly-even* asks that this half be even too:

$$n \text{ is doubly-even} \iff n = 2m \text{ with } m \text{ even.}$$

Now apply the definition of “even” a *second* time to m : m even means $m = 2k$ for some integer k . Substituting back,

$$n = 2m = 2(2k) = 4k.$$

So the doubly-even integers are precisely the **multiples of 4**:

$n \text{ is doubly-even} \iff 4 \mid n \iff n \in \{\dots, -8, -4, 0, 4, 8, 12, \dots\}.$
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1.3 Is 0 Doubly-Even?

Yes. Walk through the two requirements:

- *Is 0 even?* Yes — $0 = 2 \cdot 0$, so the guaranteed half is $m = 0$. (This is exactly the textbook’s own observation that 0 is even, with $m = 0$; the definition never required $m \neq n$.)
- *Is that $m = 0$ even?* Yes — again $0 = 2 \cdot 0$.

Both halvings land on integers, so 0 is doubly-even. (Consistently, $0 = 4 \cdot 0$ is a multiple of 4 .)

1.4 The First Three Positive Doubly-Even Integers

The positive multiples of 4, in order, are 4, 8, 12, 16, ... The first three are

4, 8, 12.

Checking each against the definition:

n	$m = n/2$	is m even?	doubly-even?
4	2	yes ($2 = 2 \cdot 1$)	yes
8	4	yes	yes
12	6	yes	yes

Table 1.1

By contrast 2, 6, 10 are even but *not* doubly-even: their halves 1, 3, 5 are odd.

1.5 Picture — Every Fourth Integer

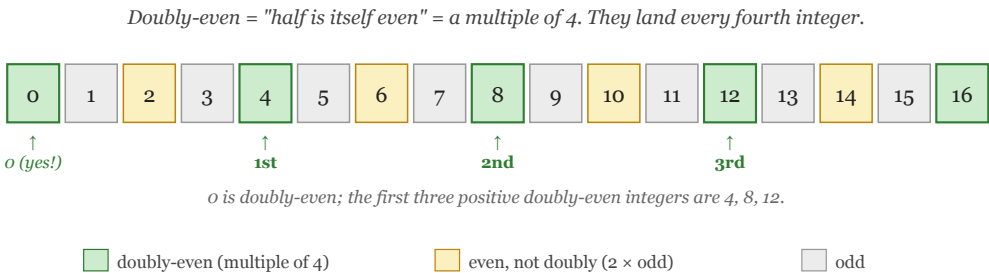


Figure 1.1: The integers 0 through 16 colored by type: odd (grey), even-but-not-doubly i.e. $2 \times \text{odd}$ (yellow: 2, 6, 10, 14), and doubly-even i.e. multiples of 4 (green: 0, 4, 8, 12, 16). Zero is doubly-even; the first three positive doubly-even integers are 4, 8, 12.

The green cells — the doubly-even integers — occur at every *fourth* position, starting from 0. Between consecutive doubly-even numbers sits exactly one “merely even” number (2, 6, 10, 14: the doubles of odd numbers) and two odd numbers.

1.6 Remark — The “Halving Test”

The definition is really a *halving test*: divide by 2 and ask whether the result is *still* even.

$$12 \xrightarrow{\div 2} 6 \text{ (even)} \implies 12 \text{ doubly-even}, \quad 6 \xrightarrow{\div 2} 3 \text{ (odd)} \implies 6 \text{ not.}$$

A number is doubly-even precisely when it survives *two* halvings and is still an integer at the end — equivalently, when 4 divides it.

1.7 Remark — Why 0 Is Special

Zero passes the doubly-even test, and in fact it passes *every* such test. The textbook flagged that “0 is even” surprises some people; doubly-evenness sharpens the point. Halving 0 gives 0 again — so 0 can be halved *forever* and never leaves the integers:

$$0 \xrightarrow{\div 2} 0 \xrightarrow{\div 2} 0 \xrightarrow{\div 2} \dots$$

So 0 is not merely doubly-even — it is “triply-even,” “quadruply-even,” and so on without limit. In the language of the next remark, 0 is *infinitely* even. The definition’s silent allowance that m may equal n (here $0 = 2 \cdot 0$) is exactly what lets 0 behave this way.

1.8 Remark — Even, but Not Doubly: the Doubles of Odd Numbers

It is worth isolating the integers that are even but *fail* to be doubly-even. These are the numbers $n = 2m$ with m *odd* — i.e.

$$n = 2 \cdot (\text{odd}) = 2, 6, 10, 14, 18, \dots$$

the integers $\equiv 2 \pmod{4}$ (yellow in the figure). Every even number is therefore exactly one of two kinds: a multiple of 4 (doubly-even) or twice an odd number (even but not doubly). This is the first step of a complete classification of integers by *how many factors of 2 they contain*.

1.9 Remark — “ k -Times Even” and the 2-adic Valuation

The pattern generalises immediately. Call n **k -times even** if it can be halved k times and stay an integer — equivalently, if $2^k \mid n$:

term	condition	examples
even	$2 \mid n$	$2, 4, 6, 8, \dots$
doubly-even	$4 \mid n$	$4, 8, 12, 16, \dots$
triply-even	$8 \mid n$	$8, 16, 24, \dots$
k -times even	$2^k \mid n$	multiples of 2^k

Table 1.2

The exact number of times a (nonzero) integer can be halved is a genuinely useful quantity in number theory: the **2-adic valuation** $v_2(n)$, the largest k with $2^k \mid n$. In these terms, *doubly-even* means $v_2(n) \geq 2$. For example $v_2(12) = 2$ (so 12 is doubly-even but not triply), while $v_2(8) = 3$ (so 8 is triply-even). And $v_2(0) = \infty$ — the formal statement of "0 is infinitely even."

1.10 Remark — The General Pattern

Putting it together, the doubly-even integers are exactly

$$\{4k : k \in \mathbb{Z}\} = \{\dots, -12, -8, -4, 0, 4, 8, 12, \dots\},$$

an evenly-spaced set hitting every fourth integer. The *positive* ones are $4, 8, 12, 16, \dots$ — the answer to the second half of the problem is their first three: $4, 8, 12$. And because the spacing is uniform, exactly one quarter of all integers are doubly-even, just as one half are even.